The Topology of Reachable Nature: Persistent Homology and the Resilience of Green-Space Access in Cities of Contrasting Morphology

Seyda Emekci
Department of Architecture, Ankara Yıldırım Beyazıt University

Abstract

Sustainable Development Goal 11.7 calls for universal access to green and public space, yet accessibility is almost always quantified as a static snapshot that says nothing about how access degrades when the street network is disrupted. We introduce a *topological resilience metric* for accessibility: the Wasserstein distance between the persistent-homology diagram of a city's green-space accessibility field before and after network disruption, traced as a function of disruption intensity ρ. The area under this degradation curve and its critical transition ρ^* summarise how robustly a place keeps its population close to nature. Applying the metric to five morphologically contrasting cities (İstanbul, Barcelona, Amsterdam, Bogotá, Phoenix) at the level of 2,603 intra-urban districts, we find that street-network morphology is a strong, statistically significant predictor of green-space access resilience: in a district-level regression with city fixed effects, street circuity, orientation entropy, mean segment length and intersection density are all significant ($p < 0.01$; three at $p < 10^{-8}$). The signal lives in dimension-0 homology—the merging of well-served regions—rather than in enclosed "access deserts," a finding that only emerges on real street networks.

1 Introduction

Target 11.7 of Sustainable Development Goal 11 commits states to provide "universal access to safe, inclusive and accessible, green and public spaces" [10]. Planners typically measure progress with catchment or proximity analyses: the share of residents within a walking threshold of a park. Such measures are static. They describe access on an ordinary day and are silent about *resilience*—how access to nature holds up when the pedestrian network is degraded by a flood, an earthquake, or the closure of a critical bridge or street. Two gaps follow. First, a city can score well on average proximity yet collapse under modest disruption; average accessibility does not reveal this fragility. Second, accessibility work rarely connects the *shape* of access to urban *form*, which is precisely the architecture–mathematics question that motivates this collection.

Persistent homology (PH) offers a principled, multi-scale, coordinate-free language for the "holes" and connected pieces of a coverage pattern [3, 7]. PH has been used to detect coverage gaps for amenities such as polling sites [6] and healthcare [5], to read the topology of urban congestion [11], and to quantify displacement under urban expansion [8]. What is *not* novel is using PH to find accessibility "holes" for an amenity. Our contribution is to move from *detecting* coverage gaps to *measuring the resilience of the access topology* and to *linking that resilience to urban form*.

Contributions.

- **A topological resilience metric** (the mathematics). We define $D(\rho)$, the Wasserstein distance between the persistent-homology diagram of the green-space accessibility field before and after disruption of intensity ρ, and summarise it by an area-under-curve (AUC) and a critical transition ρ^* (Section 4).
- **First topological treatment of SDG 11.7 green-space access** in an architecture–mathematics framing.
- **A district-level morphology↔resilience result** across 2,603 districts in five cities, with city fixed effects, showing that street-network morphology significantly predicts access resilience (Section 6).

2 Related work

TDA of spatial systems. Feng et al. [3] survey persistent homology for spatial data; Otter et al. [7] give the computational foundations. Within urban analysis, Wu et al. [11] read traffic congestion as a barcode, and a line of "resource coverage" work applies sublevel or network-distance filtrations to amenity access: polling places [6] and sexual-health services [5]. Rodríguez Vázquez and Cuerno [8] use PH to quantify displacement under urban expansion. These studies establish that PH can *locate* coverage holes. We differ on the object of study: we do not report where the deserts are, but how the whole access topology *degrades* under disruption, and we regress that degradation on morphology.

Accessibility and equity. The proximity/catchment tradition measures green-space access as a static field. Our accessibility field (Section 4) is built the same way—network walking time to the nearest green space—but is then read topologically and stress-tested, rather than thresholded once.

Street-network morphology. We characterise form with standard descriptors computed from OpenStreetMap via OSMnx [1]: intersection density, circuity, orientation entropy, and mean street-segment length. Relating these to a resilience outcome is the architecture–mathematics bridge of this paper.

3 Mathematical background

Let $G = (V, E)$ be the pedestrian network and $f : V \to \mathbb{R}$ a scalar field giving, at each node, the network walking time to the nearest green-space access point. The *sublevel-set filtration* of f builds a growing simplicial complex $K_t = \{\, \sigma : \max_{v\in\sigma} f(v) \le t \,\}$. As the threshold t (walk-minutes) rises, well-served regions appear and merge. Persistent homology records, for each topological feature, a birth b and death d; the multiset of points (b, d) is the *persistence diagram* $\mathrm{Dgm}_k(f)$ in homological dimension k. Dimension 0 tracks connected components (the merging of well-served regions); dimension 1 tracks loops (enclosed voids, i.e. access "deserts").

The distance between two diagrams is measured by an optimal matching. The 1-Wasserstein distance is

$$W_1(X, Y) = \min_{\gamma} \sum_{x\in X} \|x - \gamma(x)\|,$$

minimised over bijections γ between the diagrams (with matching to the diagonal allowed). Wasserstein and bottleneck distances are stable to perturbations of f [2], which is what makes them a sound basis for a resilience metric. We compute PH with GUDHI [9] and diagram distances with `persim`.

4 Methodology

Data. For each city we take the pedestrian network and green/public spaces from OpenStreetMap via OSMnx [1], and we fix the analysis extent to the city's *urban centre* polygon from the GHS Urban Centre Database [4] so that extents are comparable across countries. Green spaces are the OSM classes `leisure=park|garden|recreation_ground`, `landuse=grass|recreation_ground`, `place=square` and `natural=wood`; their centroids, snapped to the nearest network node, are the access points.

Accessibility field. We assign each node its network walking time (at 4.8 km/h) to the nearest green-space access point by multi-source Dijkstra, giving the field f.

Resilience metric. Let $\mathrm{Dgm}(f)$ be the baseline diagram. A disruption of intensity $\rho \in [0,1]$ removes a fraction ρ of network edges; we recompute the field and diagram on the disrupted network and define the degradation

$$D(\rho) = \mathop{\mathbb{E}}_{r} W_1\big(\mathrm{Dgm}(f),\, \mathrm{Dgm}(f_{\rho,r})\big),$$

averaged over r replicate disruptions. We consider three disruption families: uniformly random edge removal (percolation-style; used here), targeted removal of high-betweenness edges, and hazard removal of edges in low-elevation zones. The resilience of a place is summarised by the area under the curve $\mathrm{AUC} = \int_0^{\rho_{\max}} D(\rho)\, d\rho$ (normalised by the ρ-range) and the critical transition ρ^*, the smallest ρ at which $D(\rho)$ first reaches half its maximum—a percolation-style threshold at which access structure breaks down.

Homology dimension and denoising. On real street networks the sublevel filtration's finite H_1 is nearly empty: a street graph contains few triangles, so the network's cycles are never filled and almost all H_1 classes are essential (infinite) and carry no finite persistence. In Amsterdam, for example, the baseline diagram has 6,449 finite H_0 classes but only a single finite H_1 class. The resilience signal therefore lives in *dimension 0*: the merging structure of well-served regions. We consequently compute $D(\rho)$ on H_0. To keep the exact Wasserstein matching tractable on city-scale diagrams we apply persistence thresholding, discarding H_0 classes with lifetime below 1 walk-minute (topological noise); this preserves the ranking and ρ^* while reducing reported magnitudes by roughly 10–15%.

Districts and inference. Statistical inference rests not on five cities but on intra-urban *districts*: we tile each urban-centre boundary with a uniform H3 hexagonal grid (resolution 8, ≈ 0.7 km^2 cells) and compute, per district, its local resilience summaries and its local morphology (intersection density, circuity, orientation entropy, mean segment length, and a green-space fragmentation index). We then fit

$$\mathrm{AUC}_i = \beta^\top \mathbf{m}_i + \alpha_{c(i)} + \varepsilon_i,$$

an ordinary-least-squares regression of district AUC on morphology $\mathbf{m}_i$ with *city fixed effects* $\alpha_{c(i)}$ absorbing all between-city confounds. The five cities serve as a typological overlay, not as the inferential sample.

5 Case studies

We study five cities chosen to span street morphology, planning regime and Global North/South divide: İstanbul (organic, hilly, polycentric, seismic), Barcelona (Cerdà grid with superblock interventions), Amsterdam (compact, canal, walkable), Bogotá (dense informal–formal mix), and Phoenix

Table 1: District coverage (H3 resolution 8). Inference rests on $n = 2{,}603$.

City	Amsterdam	Barcelona	İstanbul	Bogotá	Phoenix	Total
Districts	227	147	729	538	962	**2,603**

Table 2: City typology (descriptive). Higher mean AUC $\Rightarrow$ access topology changes more under disruption $\Rightarrow$ *less* resilient.

City	mean AUC	mean ρ^*	mean H_0 total persistence
Amsterdam	4.25	0.27	13.11
Barcelona	6.54	0.29	13.63
Phoenix	6.58	0.18	8.08
İstanbul	7.44	0.21	12.84
Bogotá	7.54	0.24	16.90

(low-density grid sprawl). Boundaries are the GHS-UCDB urban-centre polygons; homonymous centres (e.g. Barcelona, Spain vs. Venezuela) are disambiguated by largest true (equal-area) extent.

6 Results

The random-disruption analysis yields 2,603 qualifying districts (Table 1); only 2.6% have zero AUC. District mean AUC ranks Amsterdam as the most resilient and Bogotá/İstanbul as the least (Table 2); Phoenix has the earliest breakdown ($\rho^* = 0.18$).

The headline result is the district-level regression (Table 3). With city fixed effects and $n = 2{,}603$, four of five morphology descriptors are significant, three at $p < 10^{-8}$. Because the coefficient sign is on AUC (higher AUC = less resilient), a negative coefficient means the descriptor is associated with *greater* resilience. More circuitous networks (coef -8.99), longer street segments (-0.048) and denser intersections (-0.199) are associated with more resilient green-space access, whereas higher orientation entropy ($+1.17$) is associated with less. Green-space fragmentation is not significant once it varies within cities and city fixed effects are included.

At the city scale, the Amsterdam degradation curve is monotone and smooth: $D(\rho) = (0,\ 279.5,\ 1276.1,\ 1766.1)$ at $\rho = (0, 0.1, 0.3, 0.5)$, with AUC $= 947.5$ and $\rho^* = 0.30$, confirming that the metric responds meaningfully to disruption on real data.

7 Discussion

The results give an interpretable, architecture-facing reading of resilience. Networks that are locally redundant—denser intersections, and paradoxically more circuitous routing—retain more of their well-served structure when edges are removed, because alternative short paths to green space survive. Highly ordered (low orientation-entropy) districts are, by contrast, more brittle: a regular grid offers fewer topologically distinct detours once key links fail. That the signal is carried by H_0 rather than H_1 has a concrete planning meaning: in green-dense cities the relevant structure is not isolated "deserts" surrounded by served land, but the connectivity of the served regions themselves, and it is that connectivity that disruption erodes.

Table 3: District fixed-effects regression, AUC $\sim$ morphology + C(city), n = 2,603, random-disruption scenario. Sig.: $^{***}p < 0.001$, $^{**}p < 0.01$.

Morphology descriptor	coef	std. err.	p-value
circuity	-8.991	1.185	4.5×10^{-14} ***
orientation entropy	$+1.172$	0.203	9.4×10^{-9} ***
mean street length	-0.0478	0.0059	7.2×10^{-16} ***
intersection density	-0.199	0.076	8.6×10^{-3} **
green-space fragmentation	-1.5×10^{-6}	3.8×10^{-6}	0.70

Limitations. (i) We report the random-disruption scenario; targeted (betweenness) and hazard (low-elevation) scenarios are implemented but not yet run at scale, and coefficient signs should be checked for stability across them. (ii) Results use a persistence-thresholded H_0 metric and a reduced disruption grid (5 intensities, 3 replicates); magnitudes are therefore denoised estimates, though ranking and ρ^* are stable. (iii) City-level degradation curves (Fig. 5) are so far computed for Amsterdam only. (iv) OSM completeness varies by city, and the network-distance Vietoris–Rips variant of the filtration is left to future work. None of these undermines the core finding: at the scale of thousands of districts, street-network morphology significantly predicts the resilience of green-space access.

8 Conclusion

We defined a topological resilience metric for green-space accessibility—the Wasserstein degradation curve of persistent-homology diagrams under network disruption—and showed, across 2,603 districts in five contrasting cities, that urban morphology significantly predicts how robustly a city keeps its population close to nature. The mathematics of persistence turns a static SDG 11.7 indicator into a dynamic, form-aware one. Future work will complete the targeted and hazard scenarios, add a network-metric filtration, and test resolution sensitivity, toward a resilience-aware accessibility standard for sustainable cities.

Reproducibility. All data are open (OpenStreetMap, GHS-UCDB) and the pipeline is open-source Python (OSMnx, NetworkX, GeoPandas, GUDHI/`persim`, H3, `statsmodels`); every reported number is recorded in the project's results log and traces to a committed artifact.

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